*Journal of African Earth Sciences*, Vol. 6, No. 3, pp. 353–365, 1987
Printed in Great Britain

0731-7247/87 $3.00 + 0.000
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# A paleomagnetic study of the Dokhan Volcanic Formation and Younger Granites, eastern desert of Egypt

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(*Received for publication* 19 *June* 1986)

**Abstract**—Paleomagnetic measurements on samples of late Precambrian–early Paleozoic volcanic and intrusive rocks of the Egyptian Eastern Desert show a variety of magnetic components. A stable, hematite-carried direction from the Dokhan Volcanic Formation indicates an age of about 680 Ma if the magnetization is primary and was acquired prior to structural tilting, or an age of about 500–550 Ma if it post-dates tilting. Several lines of evidence led us to prefer the latter interpretation. A high temperature component found in samples from a Dokhan xenolith within a Younger Granite pluton may also date at 500 Ma. Other directions found in the xenolith and in Younger Granites are provisionally attributed to late Mesozoic and Cenozoic overprinting.

## INTRODUCTION

INTEREST in the Arabian–Nubian Shield, now partly split by the Red Sea–Gulf of Suez, has been rekindled by the currently held view that the craton evolved from the sequential development of oceanic and island-arc environments during the interval 1100–500 Ma (Fleck *et al.* 1980, Engel *et al.* 1980). Although most of this shield is mantled by younger sediments, there are extensive outcrops of the basement rocks in Egypt on both sides of the Gulf of Suez. The outcrops in the Red Sea Hills have been known since the pioneer work of Hume (1934) and others, although with a few notable exceptions little detailed mapping or modern geophysical/geochemical surveying has been carried out. The general stratigraphy, however, has been outlined on several occasions, and Table 1 provides one of the more recent of such summaries.

The older rocks of the basement complex are commonly metamorphosed to amphibolite facies; however, the rocks of the younger formations (the Dokhan Volcanic Formation, the Hammamat Group and the Younger Granites) are not only less metamorphosed, but are lithologically suitable for paleomagnetic investigation. Their study can contribute to an understanding of the history and timing of events during the late phase of cratonic development. The present account developed from the earlier work by Davies *et al.* (1980) and Perry (1982). In the latter study, some lava xenoliths in granite had been collected, but had not yielded consistent results. During a subsequent field season, the site where the granite intrudes the Dokhan Volcanics at the western end of the Qena–Safaga road section was revisited, and an additional suite of xenolith samples and some cores in the contact granite were collected (Fig. 1). The results from this collection form the basis of this account, which also incorporates previously unpublished data on four granite sites in the same general area. Further data on the composition and tectonic setting of the Dokhan Volcanic Formation have been published by Ressetar and Monrad (1983).

## REGIONAL SETTING OF THE DOKHAN VOLCANIC FORMATION

### *Stratigraphy and age*

The type area for the Dokhan Volcanic Formation is located at Jebel Dokhan, which lies in the Eastern Desert at about latitude 27°N (Basta *et al.* 1980). The formation consists of an estimated 1200 m of extrusive, predominantly andesitic rocks with associated subordinate felsic components. The term 'Dokhan' has come to be applied to a wide compositional range of volcanic rocks showing a low grade of metamorphism and the same general stratigraphic position spread over a wide area of the Eastern Desert. While this 'practical' approach is probably valid, in the absence of detailed mapping and an extensive program of radiometric dating or paleomagnetic work, there is no guarantee that this stratigraphic assumption is correct.

Outcrops of the Dokhan Volcanic Formation are most abundant in the northern part of the Eastern Desert, and only a limited number of outcrops are known in the Red Sea Hills south of 20°N. Stratigraphic contacts with the overlying Hammamat Group and the underlying units are not common. Tectonic contacts are usually obscured by wadi fill, since the pattern of wadis is, to a large extent, controlled by faulting.

Although generally presented as stratigraphically underlying the Hammamat Group, in Wadi Quei flows of a type assigned to the Dokhan Volcanic Formation are interbedded with conglomerate and coarse

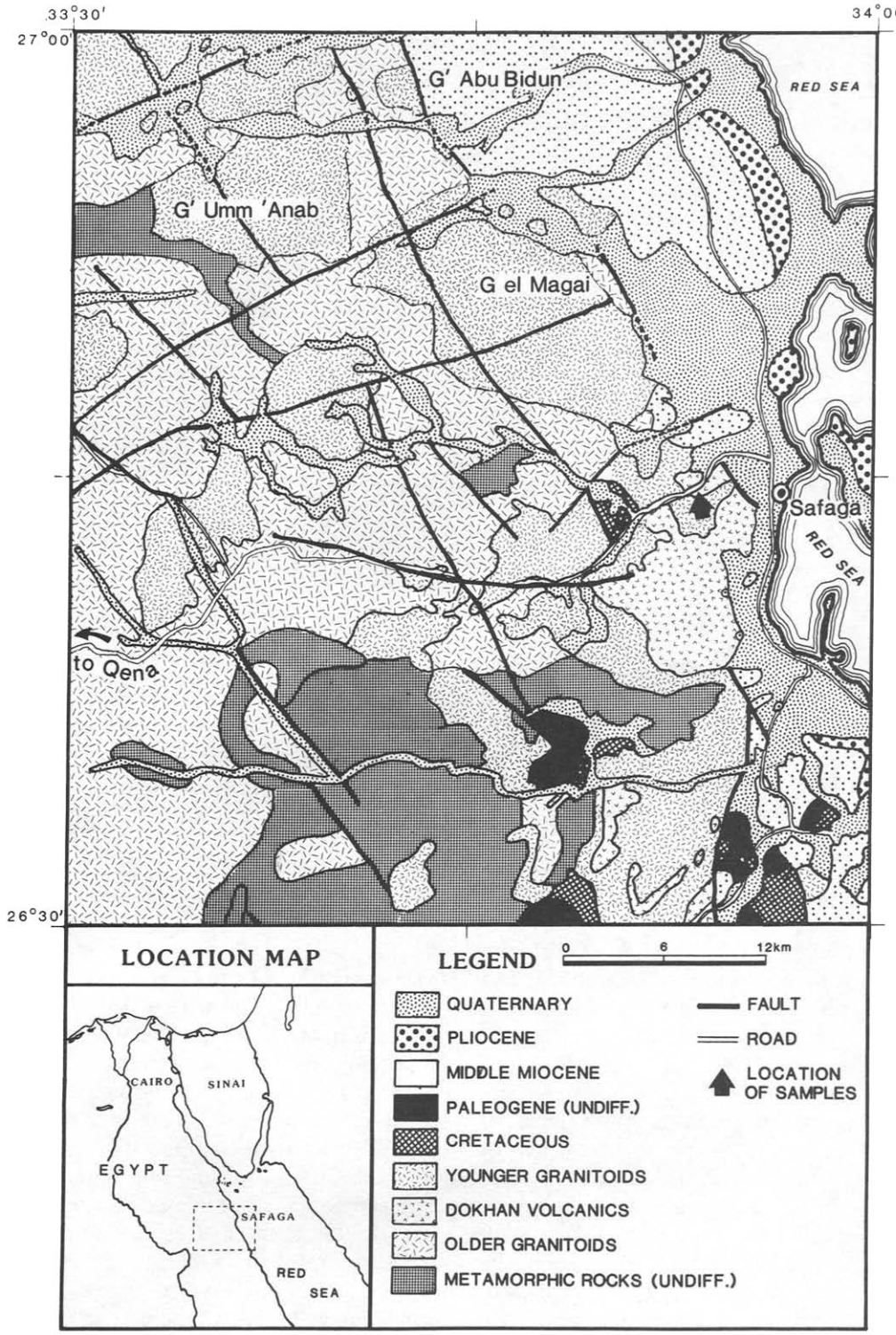



Fig. 1. Geological outline map of the Qena–Safaga road area, Eastern Desert, Egypt (after El Ramly and Hermina 1978). The regional location is shown in the insert.

sandstones, sediments of a facies regarded as belonging to the Hammamat Group. In Wadi Hamrawin, Abu Zeid (1984) divided a sequence of 'younger volcanics' into an upper and a lower series, separated by about 200 m of sandstones and conglomerates of a Hammamat facies in which some of the boulders are lithologically identical to rocks in the lower lava sequence. Although Abu Zeid (1984) regarded the elapsed time between volcanic events as significant, in the absence of unequivocal age determinations this cannot be resolved. At this stage, we prefer the interpretation that the top part of the Dokhan Volcanic Formation interfingers with, and is therefore coeval with, the lower part of the Hammamat (Ressetar and Monrad 1983, Stern *et al.* 1984).

Several radiometric age determinations are available for the Dokhan Volcanic Formation. Stern and Hedge (1985) reported an Rb/Sr age of 592 ± 13 Ma for flows at the type locality. At the Qena–Safaga road site, they obtained isochrons of 581 ± 7 Ma and 686 ± 28 Ma for

Table 1. Simplified stratigraphic table of the basement complex in the Red Sea Hills, Eastern Desert of Egypt, after Ressetar and Monrad (1983), drawn largely from El Shazly (1977)

| Unit name | Rb/S and/or U/Pb Age (Ma) | Inferred Tectonic Setting | References |
|---|---|---|---|
| Younger Granites | 606–580 | Young, stable craton | Fullagar and Greenberg 1978, Greenberg 1981 |
| Hammamat Group | — | Young, block-faulted craton | Grothaus *et al.* 1979 |
| Dokhan Formation | 620–560 | Young, block-faulted craton | Stern 1979, Hurst pers. comm. and Table 5 |
| Older Granites | 711–610 | Island arc | Dixon 1981, Stern 1981 |
| Younger Metavolcanics | 761–612 | Island arc | Stern 1979 |
| Metasediments | — | Island arc; oceanic crust | Engel *et al.* 1980 |
| Older Metavolcanics, Metagabbros and Serpentinites | 1100 | Oceanic crust | Fleck *et al.* 1980 |

The ages shown are those most recently available in the literature.
The inferred tectonic settings are based on the references given in the title and in the table.

rhyolitic and andesitic rocks, respectively, but regarded the older age as erroneous. The oldest age, therefore, overlaps the ages assigned to the calc-alkali island arc metavolcanic assemblage of Stern (1979). The younger ages overlap the age range assigned to the post-tectonic younger granites (Fullagar and Greenberg 1978).

Therefore, although there are considerable uncertainties about the precise age of the rocks of the Dokhan Volcanic Formation, most data are consistent with their extrusion during the waning phases of the Pan-African event (Kennedy 1964) in a tectonic environment transitional to a stable craton during latest Precambrian or early Phanerozoic time. As the best age estimate of Dokhan activity, the date of approximately 590 Ma suggested by Stern and Hedge (1985) is adopted. During the latter part of this activity, at least, the lavas interfingered with sediments deposited in a horst and graben extensional regime (Grothaus *et al.* 1979).

### *Petrographic character*

For the type locality at Gebel Dokhan, Basta *et al.* (1978) described a sequence of predominantly porphyritic andesites, including the historically important decorative stone, the Imperial Porphyry, and subordinate, more felsic, components. In Wadi Hamrawin, Abu Zeid (1984) recognized ignimbrites and rhyolites among the more felsic components. In the Qena–Safaga road section, there is a sequence of porphyritic andesites in which are interbedded dacite, trachyte, and rhyolite flows, as well as tuff.

The andesites and dacites are generally porphyritic, the most common phenocrystic material being euhedral to subhedral, sodic to intermediate plagioclase, which can reach 1 cm in size, although crystals are usually much smaller. Occasionally, a twinned (albite) phenocryst may have an untwinned rim. In the rhyolites, sanidine is the principal feldspar phenocryst phase, and may commonly be accompanied by embayed quartz crystals. Hornblende phenocrysts are surprisingly rare in the andesites as well as the rhyolites.

The groundmass usually consists of albite-twinned feldspar laths, with abundant opaque minerals and biotite; hornblende is found in the more acidic varieties. The texture is usually intergranular or trachytic; devitrification textures are common in the more acid rocks.

Opaque oxides are conspicuous both in phenocrysts and in the groundmass. Both magnetite and hematite occur—the latter being more common in the rhyolitic rocks where it is present as a fine powder and either disseminated or aggregated in clumps with magnetite. Magnetite is quite common in the andesites, and phenocrysts typically show ilmenite exsolution lamellae.

The common alteration products are sausserite and sericite, seen in the interiors of some phenocrysts. Uralite aggregates are common in the groundmass in the more acidic rocks. Vesicles, when found, contain calcite, chlorite and prehnite. This mineral assemblage suggests that the degree of alteration is consistent with low grade, greenschist grade, hydrothermal metamorphism.

## PALEOMAGNETIC STUDY

### *Techniques of study*

Oriented blocks and cores drilled out with a water-cooled drill from the lavas in this section were both oriented with a Brunton compass. In addition to the flows of the Dokhan Volcanic Formation, samples of the lava xenoliths and the contact granite were collected. Granite samples had previously been collected at four outcrops by Sedivy (unpublished data). Each flow and granite outcrop was regarded as a site, and up to six oriented cores or block samples were taken. This system could not be applied to the xenoliths and contact granite samples, which are treated as a single group.

The natural remanent magnetization (NRM) of all samples was measured using a Digico complete-results magnetometer interfaced with a MINC II microcomputer. A few samples were progressively demagnetized in alternating fields up to 100 mT, the limit of the Schonstedt AC demagnetizer. Figure 2a gives examples of two of the samples so examined, showing that there was very little response. More satisfactory results were arrived at using the Schonstedt thermal demagnetizer; Figs. 2b, c and d give examples of the varied response. The degree of overlap of the coercivity spectra of the magnetic component is variable, and in some cases no discrimination between the different components is possible. The characteristic spectra of the lava samples were

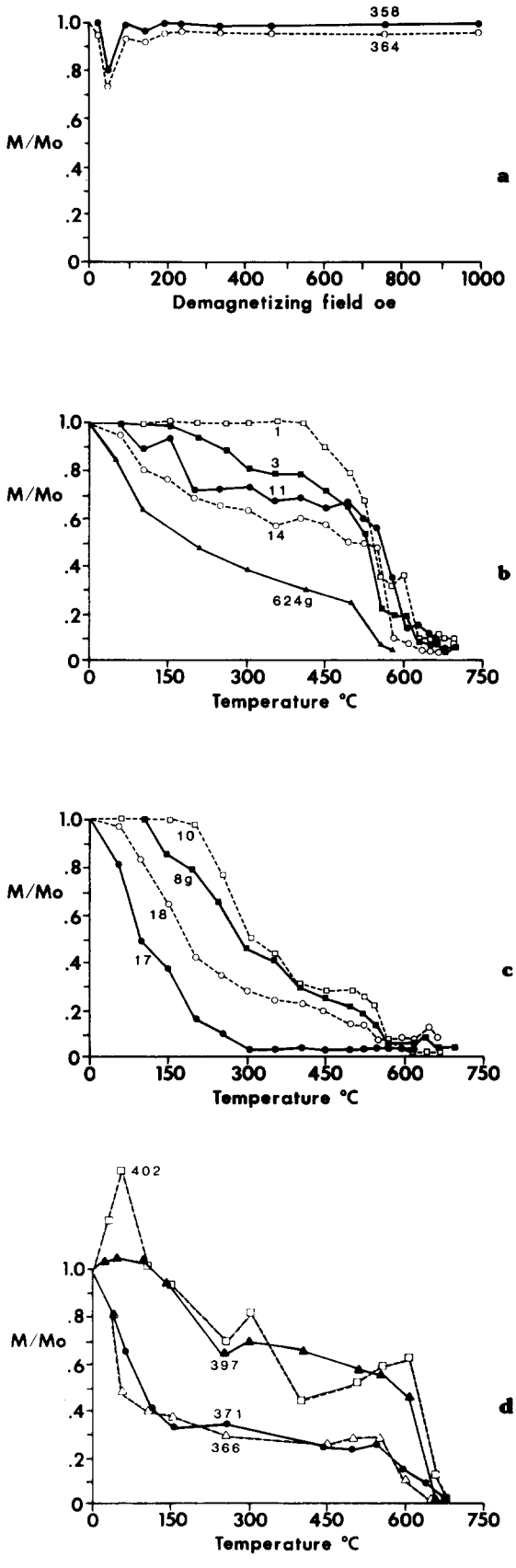



Fig. 2. Demagnetization curves for samples from the Qena–Safaga road section. Vertical axis (M/Mo) is intensity of magnetization relative to the intensity of the natural remanent magnetization. Horizontal axis is the demagnetizing temperature or field strength. (a) Alternating field demagnetization of Dokhan volcanic samples; (b, c) thermal demagnetization of xenolith and granite (8 and 624 g) samples; (d) thermal demagnetization of Dokhan volcanic samples.

not normally resolved until treatment at 500°C or above (see Perry 1982). Batch techniques were not satisfactory, and essentially all samples (lava, xenolith and granite) were demagnetized at 350°C, in 50°C steps until 500°C, at 25°C intervals to 650°C, then at 660, 665 and 675°C. Site mean data are summarized in Tables 2, 3 and 4.

Vector removal techniques were then applied. The vector removed between individual demagnetization stages tended to be scattered, perhaps as a result of the generally low intensity between closely spaced steps. These are further discussed in the following section.

### *Dokhan Volcanic Formation samples*

The characteristic declination and inclination for most sites was recovered following thermal demagnetization between 500 and 675°C, although, as is apparent from inspection of Table 2, usually at least one sample per site did not respond. Data were too dispersed in six of the 16 sites, and these were omitted from calculations, although individual samples within each of these sites could be found with directions of magnetization consistent with those of the remaining 10 sites. The site mean directions of the sites are depicted in Fig. 3, and, together with the appropriate statistical parameters, are listed in Table 2 (taken from Perry 1982).

The thermal demagnetization curves (Fig. 2d) show a range of blocking temperatures indicating that remanence is carried by both titano-magnetite and hematite. From the stable endpoint curves (Fig. 4) and stereograms (Fig. 5) it appears that the dominant direction of remanence is the same. This direction has a southerly declination and a moderate positive inclination; in many cases, it is persistent up to the Curie temperature of hematite (e.g. sample AD389). The results of magnetic cleaning of three sites are illustrated by the stereogram in Fig. 6, which gives the results from individual samples.

There was evidence of the existence of other secondary components of magnetization, but these were only isolated in two sites (Table 2). In the other sites, although carried by an occasional sample, a systematic overprint could not be isolated.

### *The granite samples*

Measurements were available for granites collected several years previously from four sites in the vicinity of the Qena–Safaga road, well away from the Dokhan section. These form part of the suite of Younger Pink Granites mapped as continuous with the granite whose margins were sampled in this study. The intensity of magnetization lies within the range found in the contact granite, and their demagnetization is similar (Fig. 2 curve, 624 g). The site mean directions of magnetization isolated at high temperatures (see Table 3) indicate that both normal and reversed directions were recorded. Unfortunately, vector removed techniques were not applied to these samples.

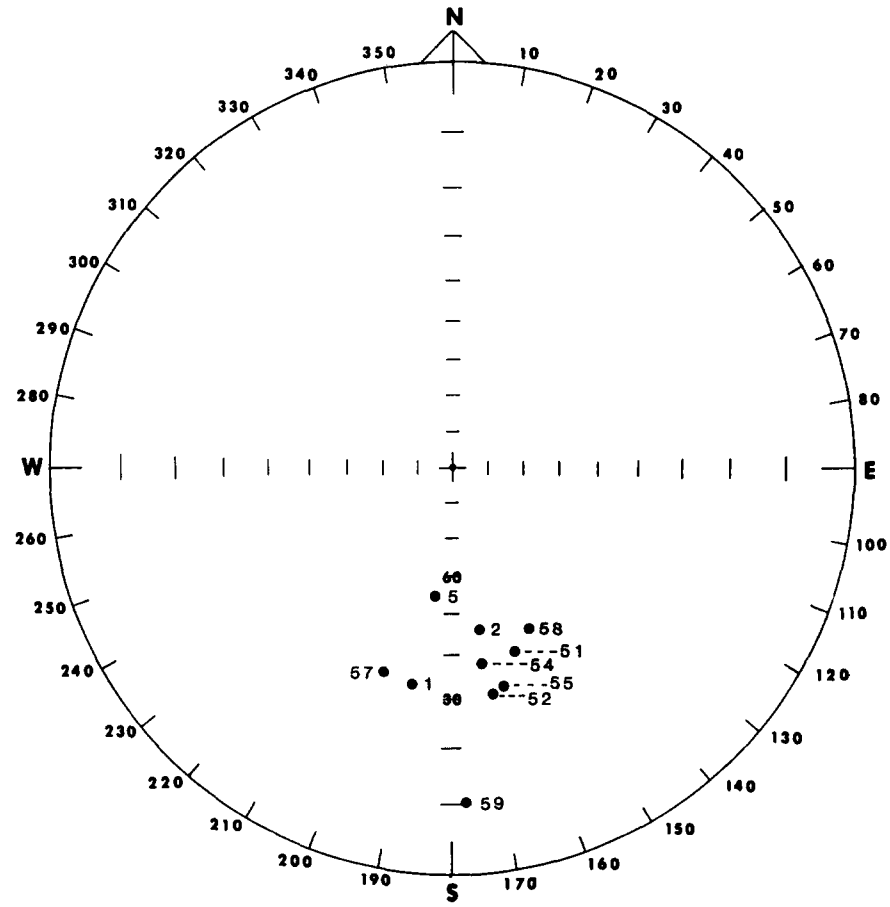



Fig. 3. Stereogram of site mean directions of magnetization of lava samples from the Dokhan Volcanic Formation, Qena–Safaga road. On this and other stereograms, closed symbols show positive inclinations and open symbols show negative inclinations.

Table 2. Summary of site mean directions of magnetization and statistical parameters of lavas from the Dokhan Volcanic Formation from the Qena–Safaga Road (after Perry 1982), after correction for geologic dip. The data for sites 61 and 62 are high temperature components. Sites 1–5 are from Davis *et al.* (1980)

| Site | Declination | Inclination | $\alpha_{95}$ | $K$ | $R$ | Demag stage (°C) | Samples measured/collected |
|---|---|---|---|---|---|---|---|
| 51 | 165.5 | 39.1 | 9.1 | 27.0 | 4.94 | 675 | 5/7 |
| 52 | 171.6 | 31.2 | 10.8 | 50.0 | 5.90 | 500 | 6/6 |
| 53 | 187.0 | 22.1 | 74.8 | 2.5 | 2.79 | 500 | 4/5 |
| 54 | 174.8 | 37.8 | 21.4 | 19.0 | 3.84 | 450 | 4/5 |
| 55 | 168.9 | 30.6 | 10.44 | 42.0 | 5.88 | 450 | 6/6 |
| 56 | 5.5 | −0.12 | 13.6 | 46.0 | 3.93 | 650 | 4/7 |
| 57 | 201.4 | 34.0 | 33.8 | 6.0 | 4.34 | 450 | 5/6 |
| 58 | 158.9 | 42.2 | 14.6 | 40.0 | 3.92 | 500 | 4/5 |
| 59 | 178.3 | 10.4 | 34.4 | 8.0 | 4.60 | 675 | 4/8 |
| 60 | 265.3 | 51.3 | 44.1 | 9.0 | 2.77 | 500 | 3/5 |
| 61 | 300.9 | −8.1 | 22.9 | 12.0 | 4.66 | 600 | 5/5 |
| 62 | 313.0 | −39.0 | 18.4 | 26.0 | 3.88 | 650 | 4/6 |
| 1 | 192.8 | 33.4 | 15.7 | 24.5 | 4.84 | 500 | 5/6 |
| 2 | 176.8 | 47.1 | 14.7 | 27.5 | 4.86 | 600 | 5/6 |
| 3 | 167.6 | 36.0 | 90.0 | 1.3 | — | 500 | 5/6 |
| 5 | 195.4 | 55.6 | 9.3 | 67.9 | 4.94 | 600 | 5/6 |
| Mean* | 178.1 | 36.8 | 9.73 | 33.3 | 9.64 | | 10/16 |
| Pole position: latitude = 42.7S, longitude = 36.2E | | | | | | | |

* Mean excludes sites 53, 56, 60, 61, 62, 3.

Column headings: site, mean declination and inclination; $\alpha_{95}$, semi-angle of the cone of confidence; $K$, a measure of dispersion; $R$, length of the mean vector; demagnetization stage at which the characteristic magnetization was calculated; number of samples used in the mean calculation and total of samples collected.

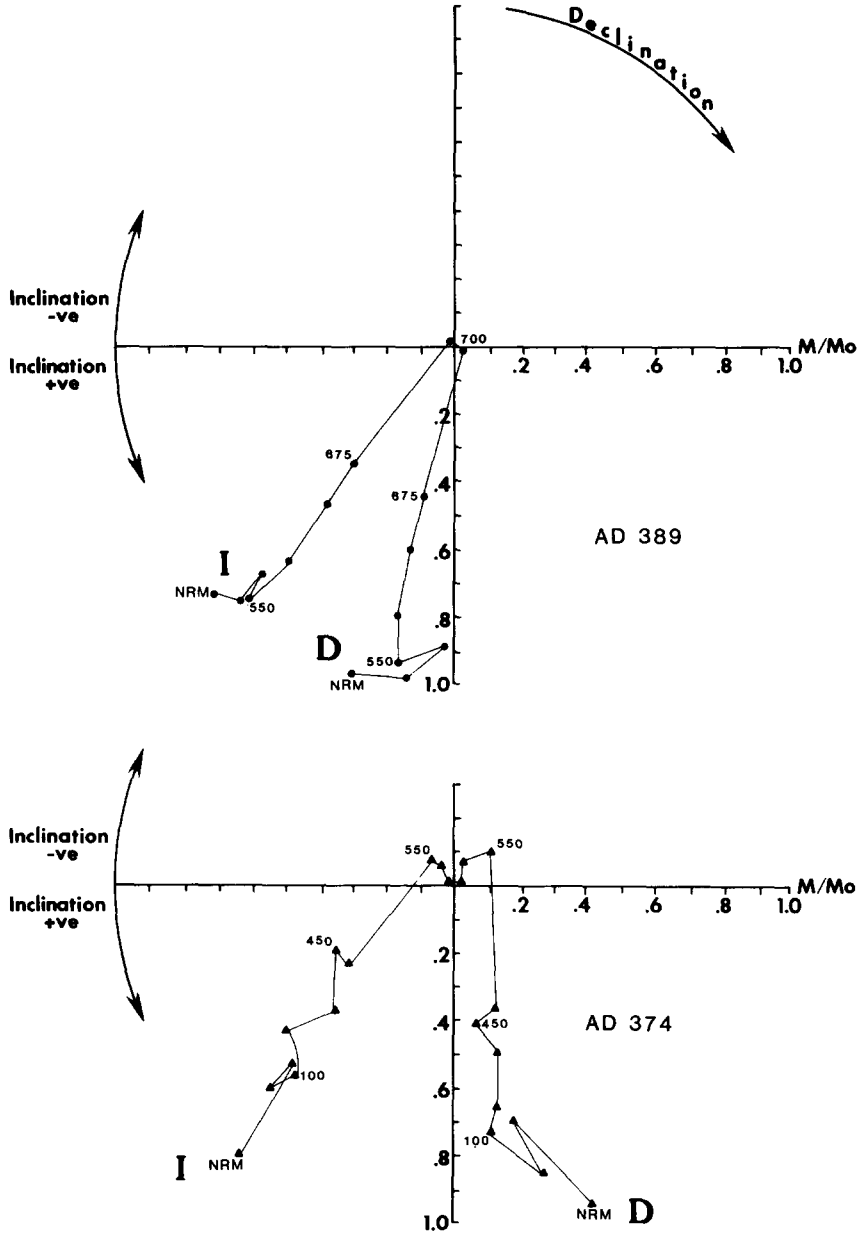



Fig. 4. Normalized stable endpoint curves of two samples from the Dokhan Volcanic Formation. These curves show plots of the normalized intensity of magnetization (M/Mo measured by distance from the origin) against the declination (D) in the x–y plane and inclination (I) in degrees from the horizontal in the vertical plane.

Table 3. Summary of site mean directions of magnetization and statistical parameters of sites in the Younger Pink Granites from the Qena–Safaga region

| Site | Declination | Inclination | $\alpha_{95}$ | $K$ | $R$ | Demag stage (°C) | Samples measured/collected |
|---|---|---|---|---|---|---|---|
| 1 | 177.4 | −20.0 | 39.2 | 6.0 | 2.535 | 575 | 3/5 |
| 2 | 178.1 | −53.2 | 13.5 | 33.0 | 4.878 | 575 | 5/6 |
| 3 | 180.0 | −35.1 | 29.1 | 24.0 | 4.489 | 575 | 5/6 |
| 4 | 347.8 | 37.6 | 16.3 | 32.0 | 3.907 | 550–575 | 4/4 |
| Mean | 175.9 | −36.5 | 16.4 | 22.0 | 3.907 | | |
| Pole position: latitude = 82.5S, longitude = 65E | | | | | | | |

Column headings: site, mean declination and inclination; $\alpha_{95}$, semi-angle of the cone of confidence; $K$, a measure of dispersion; $R$, length of the mean vector; demagnetization stage at which the characteristic magnetization was calculated; number of samples used in the mean calculation and total of samples collected.

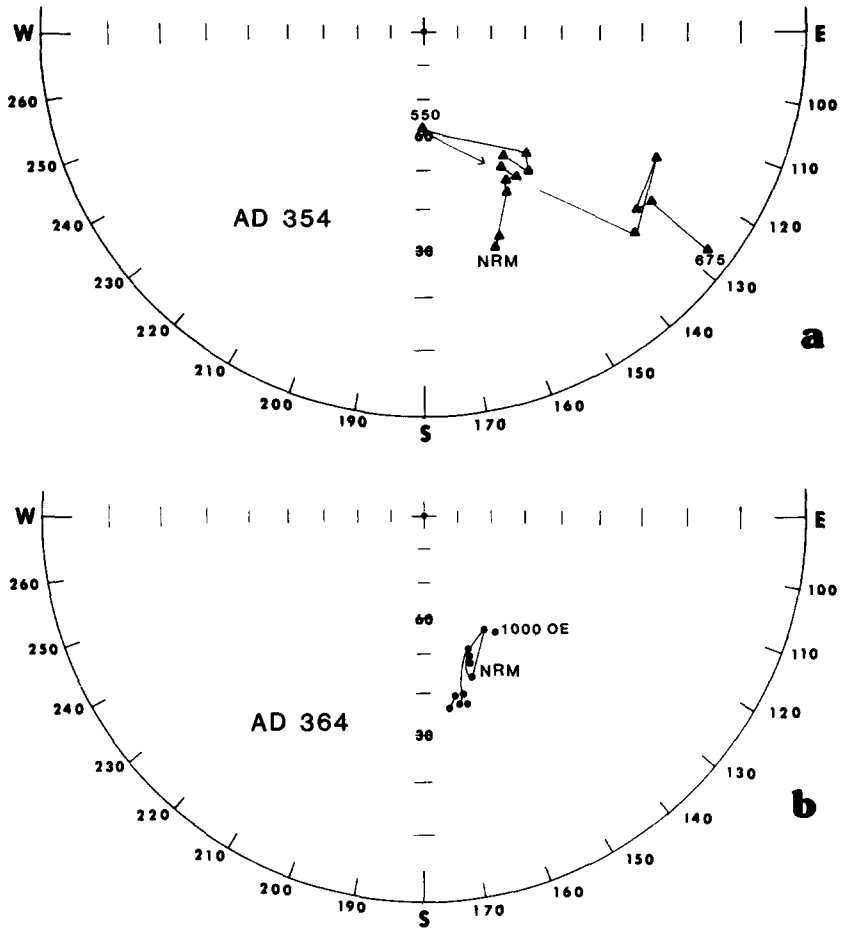



Fig. 5. Stereogram showing changes in the directions of magnetization of two Dokhan lava samples: (a) as a result of thermal demagnetization and (b) as a result of demagnetization in alternating fields.

Table 4. Summary of the magnetic components isolated in xenolith and contact granite samples from the west end of the Dokhan Volcanic Formation outcrop along the Qena–Safaga Road, Eastern Desert, Egypt

| Vector | Temperature range (°C) | Samples | Declination | Inclination | $\alpha_{95}$ | $K$ | $R$ |
|---|---|---|---|---|---|---|---|
| A | at 660 | 10, 10a, 11, 12, 13, 14, 16, 17, 18, 19 | 187.7 | −18.7 | 15.1 | | 9.194 |
| A′ | at 660 | 10 | 170.0 | −20.0 | — | | — |
| B | at 660–665 | 2, 3, 4, 5, 6 | 358.2 | 7.0 | 16.8 | | 4.8 |
| B′ | betw 450–500 | 1, 3, 7, 8 | 4.0 | 9.4 | 27.0 | | 3.8 |
| Mean A, A′–B, B′ | | | 359.8 | 13.8 | 11.1 | | 3.957 |
| Pole position: longitude = 145.5W, latitude = 70.3N | | | | | | | |
| C | at 150 | 1, 17, 18, 19 | 352.0 | 59.3 | 10.9 | | 3.8 |
| C′ | above 350 | 10a, 17, 18, 19 | 355.6 | 48.8 | 14.3 | | 3.9 |
| Mean C, C′ | | | 354.0 | 53.9 | — | | — |
| Pole position: longitude = 1.5E, latitude = 80.7N | | | | | | | |
| D | up to 500 | 11, 12, 13, 14, 16 | 234.5 | 50.8 | 5.3 | 211 | 4.981 |
| D′ | up to 670 | 5, 10a, 13 | 231.6 | 56.4 | 10.8 | 131 | 2.985 |
| Mean D, D′ | | | 233.2 | 53.6 | 12.8 | | |
| Pole position: longitude = 86W, latitude = 10.8S | | | | | | | |
| Mean D, D′ corrected for geological dip | | | 194.0 | 36.0 | | | |
| Pole position: longitude = 164W, latitude = 41N | | | | | | | |
| 6E 41S | | | | | | | |
| E | at 660 | 7, 8, 9 | 40.1 | −24.9 | 39.9 | | 2.811 |
| Pole position: longitude = 164E, latitude = 34N | | | | | | | |

All measurements are made with respect to the present surface.

Column headings: mean declination and inclination; $\alpha_{95}$, semi-angle of the cone of confidence; $K$, a measure of dispersion; $R$, length of the mean vector.

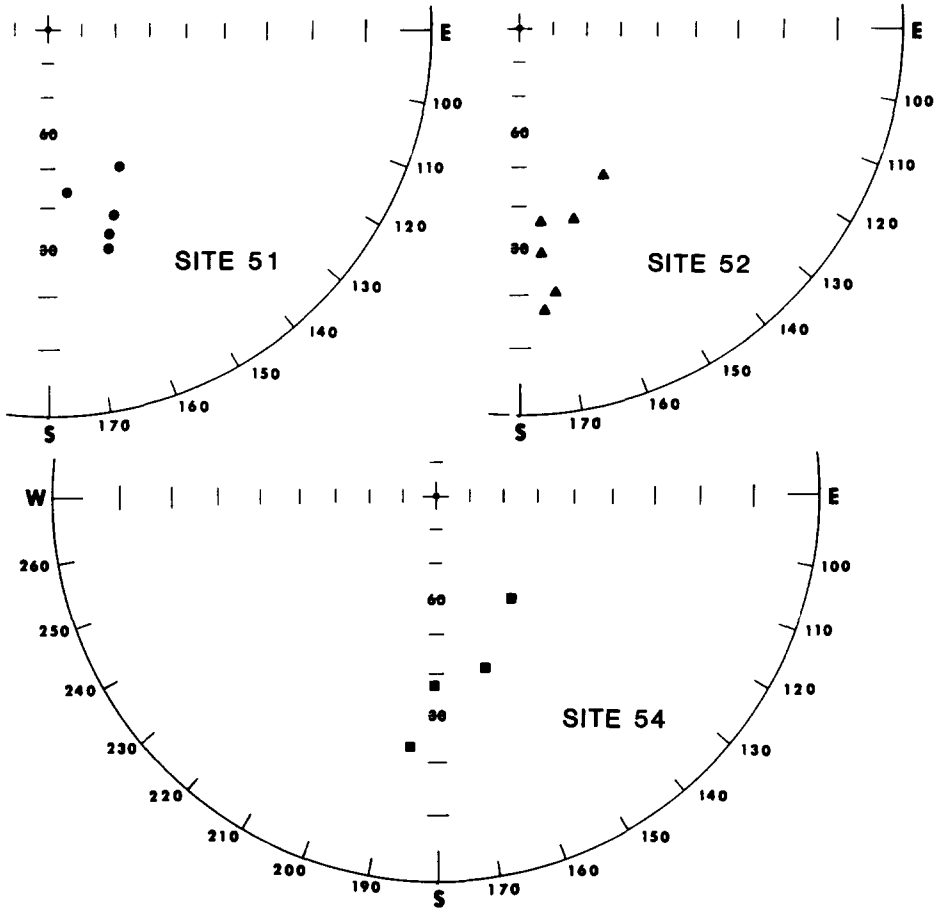



Fig. 6. Stereogram of the cleaned directions of magnetization of lava samples from sites 51, 52 and 54 from the Dokhan Volcanic Formation.

*Contact granite and lava xenolith samples*

The normalized demagnetization curves (Fig. 2) show the variable response to thermal demagnetization of both the xenolith and lava samples. Essentially all of them show an abrupt reduction in intensity of magnetization above 550°C, but a weak magnetic component persists at temperatures well over 600°C. This is consistent with the recognition among the opaque minerals in thin section of both altered magnetite and hematite.

The magnetization of the xenolith and contact granite samples is complex and it is necessary to consider both the removed and residual magnetic vectors. The residual magnetic components are given letter designations, while the vector removed to reach that direction is indicated by a prime (e.g. A and A'). Only one of the 18 samples examined proved to have a single magnetic component after the removal of a viscous component at low temperatures. The results from this sample are illustrated in Fig. 7. The stable endpoint curve shows a linear decrease to the origin above 200°C. The uniformity of the directions of magnetization can be more clearly represented by stereograms, where in one the residual direction is plotted (Fig. 7b), and in the other, the declination and inclination of the removed vector is plotted (Fig. 7c). These are designated the A and A' directions, respectively (see Table 4).

The A direction is found as the high temperature residual component in 10 of the samples examined (Table 4 provides a summary of the components identified), but it is not found in any other samples as the vector removed (A').

Table 5. K/Ar minimum ages of five samples from the Dokhan Volcanic Formation, Qena–Safaga Road, Eastern Desert of Egypt

| Sample name | Sample mass | Potassium (%) | Radiogenic (%) | Argon (nmol $g^{-1}$) | Calculated age (Ma)* | Description |
|---|---|---|---|---|---|---|
| AD 345 | 0.1592 | 0.944 | 91.42 | 1.0375 | 543.351 | Andesine phenocrysts |
| AD 351 | 0.2710 | 0.959 | 91.07 | 0.9874 | 513.492 | Fine grained, mafic |
| AD 353 | 0.1528 | 1.713 | 97.12 | 1.8408 | 532.719 | Fine grained, mafic |
| AD 376 | 0.1724 | 1.000 | 92.61 | 1.1513 | 565.347 | Andesine phenocrysts |
| AD 379 | 0.1001 | 0.756 | 83.69 | 0.6835 | 457.988 | Andesine phenocrysts |

* Based on decay constants in Jaeger *et al.* 1977.

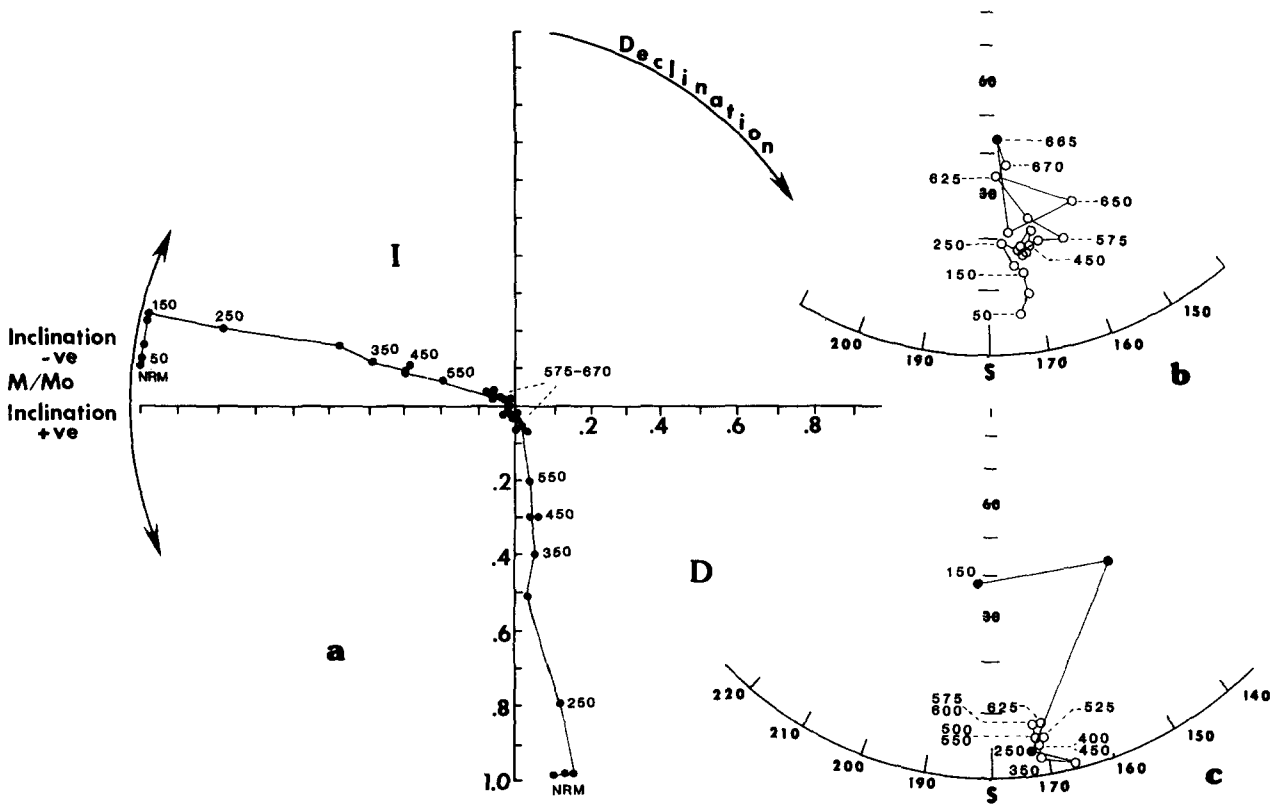



Fig. 7. (a) Shows stable endpoint curve from lava xenolith sample 10. This is the only single component sample (after the removal of a small viscous component), as shown by the fact that the direction of magnetization of the residual vector A (stereogram (b)) and that of the removed vector A′ (stereogram (c)) remain the same.

A B component, antipodal to the A direction, is found in all the remaining samples. The example in Figs. 8a and 9a shows the B (residual) component dominant in the stable endpoint curve, but as the stereogram (Fig. 8b) shows, there appears to be a further component, E, in the contact granite samples above 500°C, although declination and inclination of the vector removed (Fig. 8c) over most of the range is B′. The B component is the high temperature component in the xenolith samples, but disappears at about 550°C in the granite samples.

In many samples, the dominant magnetization in the lower temperature range is a steeply dipping, northerly component, C (Fig. 10a,b), which can also be found as the vector removed C′ at higher temperatures (Fig. 10c). No antipodal to this direction has been found.

The only component found in some of the xenolith samples that was not found in the contact granite samples is component D (Fig. 11a), the residual vector seen up to 500°C, and occurring as the vector removed D′ up to 670°C in three samples.

## DISCUSSION

Geometric pole positions for the lavas, granites and xenoliths have been plotted along with earlier results from dike studies from the Red Sea Hills and are shown in Fig. 12. The complexity of the structural and dating problems do not permit a unique interpretation of these data, but some possible interpretations are presented for they provide valuable constraints and point to directions for further research. These interpretations, however, ignore possible tilting imposed by the Cenozoic rifting of the Red Sea–Gulf of Suez region. In some places the plane of the Nubian unconformity can be seen to be essentially flat-lying, but on some basement blocks near the Red Sea the basal Miocene unconformity may show a tilt of 10–15°. While reducing the value of the magnetic data, tilt corrections of the order of 10–15° do not obscure the two interpretations proposed.

The first point to be considered is whether the secondary magnetization of the Dokhan Volcanics was acquired prior to or post-tilting. In Fig. 12, the geomagnetic pole positions for both cases are plotted on the McWilliams and Kroner (1981) curve. Pole DV-1 is the pole position after the dip correction, i.e. it assumes magnetization was acquired before tilting, while DV-2 is the post-tilt magnetization pole. If ages can be assumed from the curve, then dates of over 500 and over 600 Ma are implied. Stern and Hedge (1985) find two possible ages for the Dokhan rocks of the Qena–Safaga road: one in the age range 581 ± 7 Ma, for rhyolitic varieties, and the other 686 ± 26 Ma, for the lower andesites and dikes. The apparently conformable relationship between the andesites and rhyolites of the Dokhan Volcanic Formation led Stern and Hedge (1985) to reject the older age in their interpretation of the Rb/Sr data. If they are correct, then this would eliminate a pre-tilt magnetization of the Dokhan Volcanics because of the location of the pre-tilt pole on the McWilliams and Kroner curve.

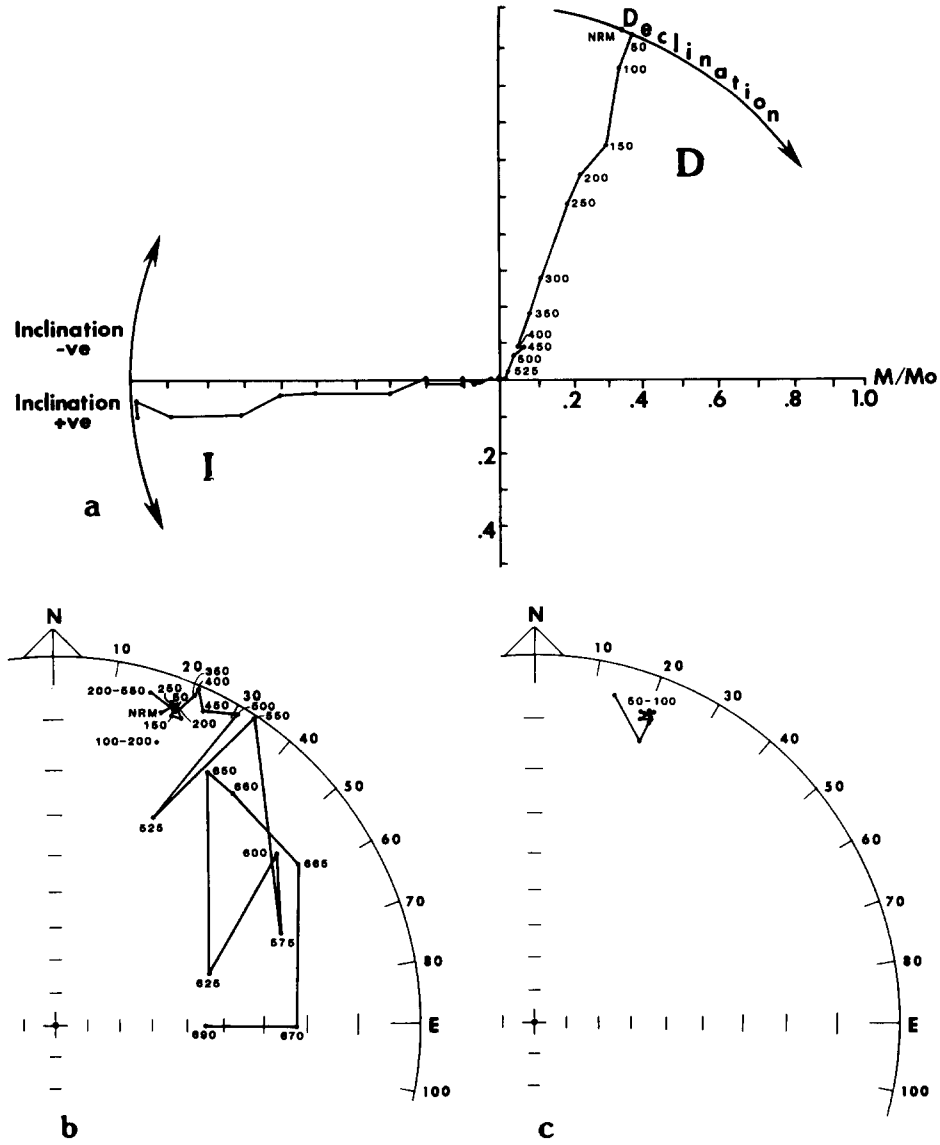



Fig. 8. (a) Shows stable endpoint curve from contact granite sample 7, showing the declination and inclination of the dominant component B up to 450°C (stereogram (b)). This is also the declination and inclination of the component removed B′ (stereogram (c)). At temperatures above 550°C, there appears to be a poorly defined component, E, with a northeast, positive direction.

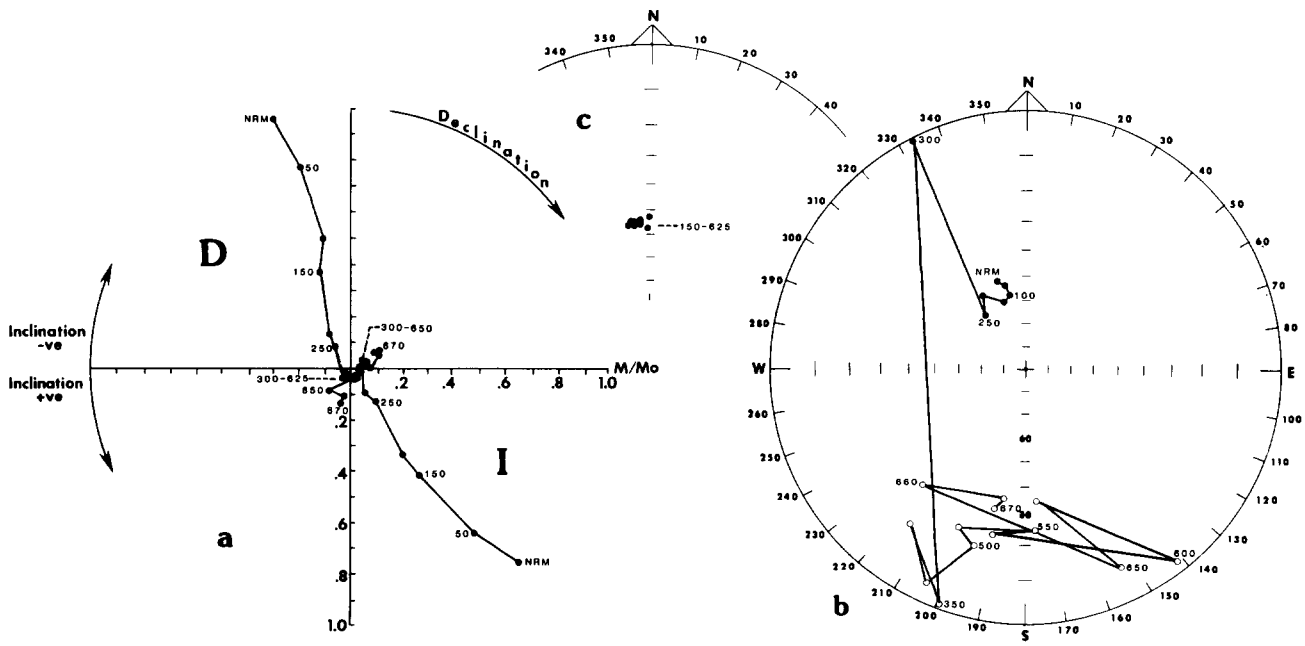



Fig. 9. (a) Shows stable endpoint curve from lava xenolith sample 17. The declination and inclination of component C is dominant up to 250°C as is shown in stereogram (b), while above 350°C the direction of magnetization of component A is recovered with continuing removal of a component with the declination and inclination of C′ (stereogram (c)).

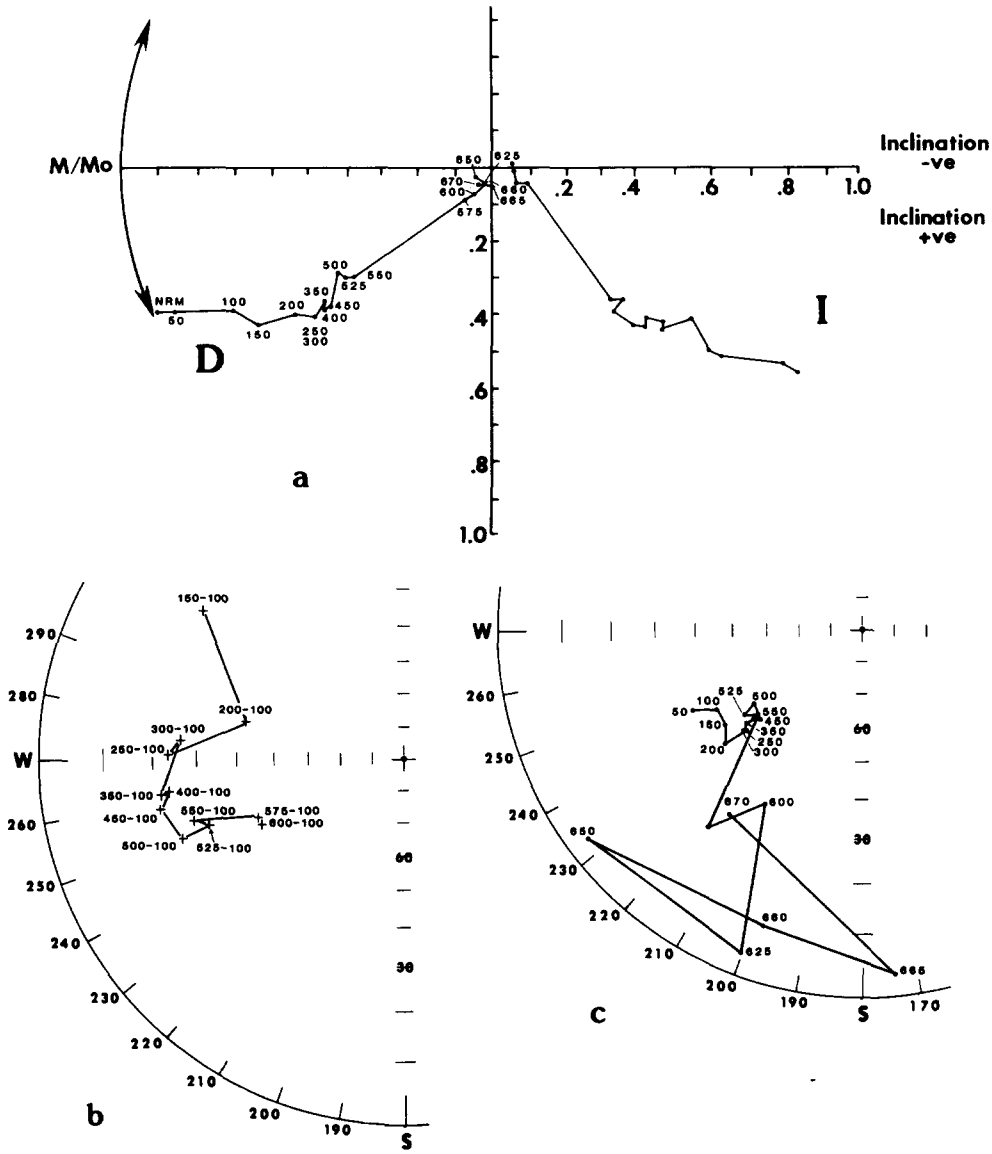



Fig. 10. (a) Shows stable endpoint curve from lava xenolith sample 14. The residual vector D persists up to 550°C (stereogram (c)) close to the direction of the vector removed D′ (stereogram (b)). A complete isolation of a single residual vector is not possible, but above 620°C component A clearly is present.

We accept this interpretation; the magnetizations recorded in the rhyolites and andesites are similar and imply no major break in extrusive activity. The interpretation is consistent with K/Ar minimum ages in the range 480–565 Ma (Table 5) found in a variety of igneous rocks in the Eastern Desert, including rocks of the Dokhan Volcanic Formation (Davies *et al.* 1980, Perry 1982, Abdullah *et al.* 1984). It also avoids an alternative interpretation which would require that the magnetization of the xenoliths was acquired without any rotation from their original position during granite emplacement to explain the D component.

There is a greater degree of uncertainty in the interpretation of the poles derived from the magnetization of the Younger Granites and from the AB and C components derived from the Dokhan Volcanic xenoliths. These poles lie at high latitudes, grouping with poles derived from early Paleozoic (post-Dokhan) basic dike swarms in the Eastern Desert (poles QS, UR and EEM in Fig. 12). Their proximity to the present pole immediately suggests a late Phanerozoic age for these secondary magnetizations. This is constrained by the argument for relatively high latitudes for the African landmass during the Paleozoic to be consistent with the occurrence of continental glaciations during the Lower (Late Ordovician to Early Silurian) and Upper (Dwyka) Paleozoic in both northern and southern Africa—an argument which is only partly supported by paleomagnetic data. The question this interpretation leaves unanswered is why the AB and C components so far have only been found in the xenoliths of the Dokhan Volcanics and not in 'normal' Dokhan samples.

The C component, a lower coercivity component of the contact rocks, fits within the grouping of magnetizations from the dike swarms (see Fig. 12). The pole of the AB components, a magnetization presumably carried by hematite (see Table 4), on the other hand, groups with the pole derived from the Cretaceous igneous rocks of the Eastern Desert (Ressetar *et al.* 1981).

In summary, the magnetic components isolated from

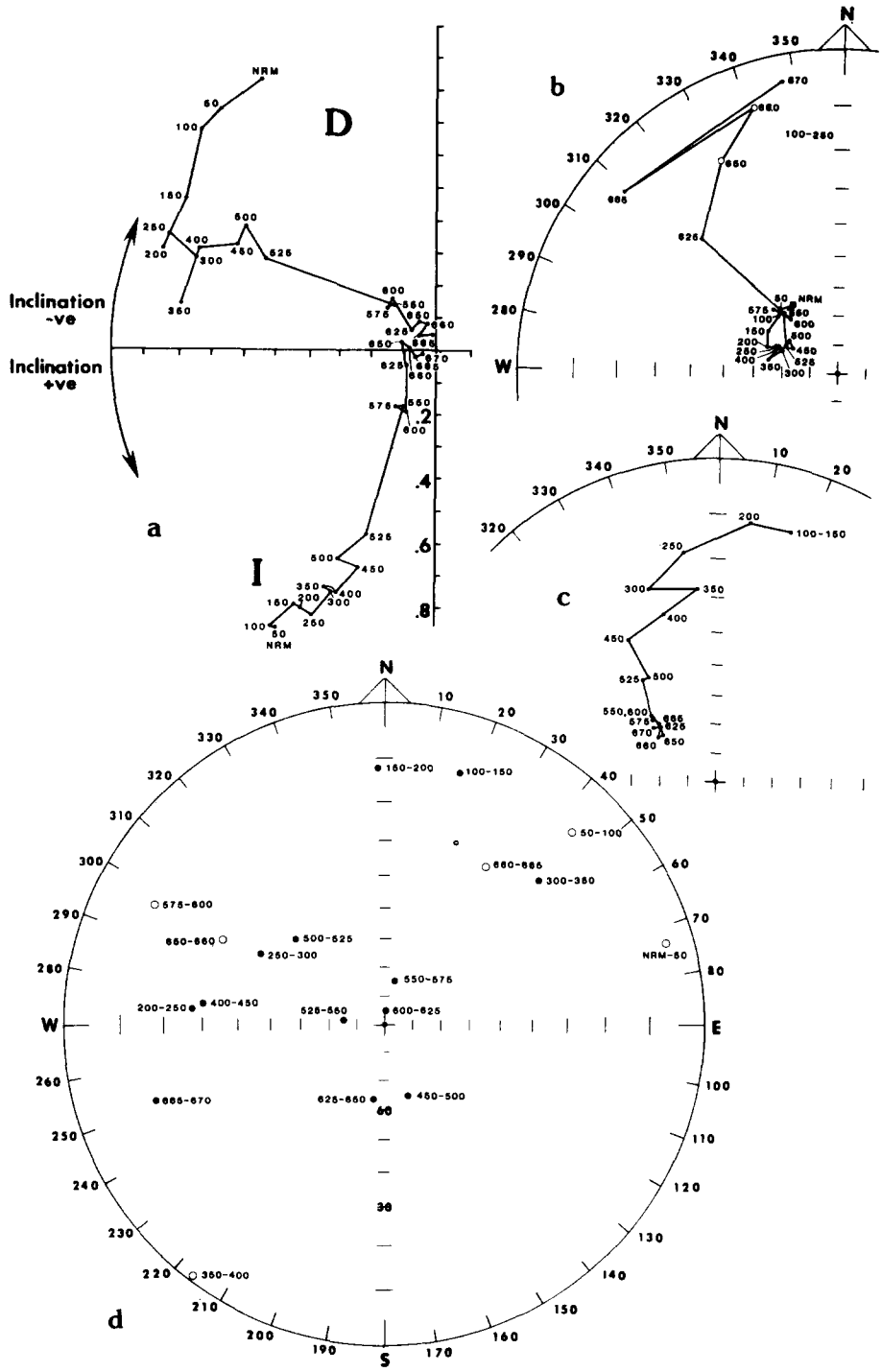



Fig. 11. (a) Shows stable endpoint curve for lava xenolith sample 3, which shows initially component C) at high temperatures, the residual component approaches component B (stereogram (b)). The vector removed shows a progressive change and only at high temperatures appears to indicate component C′ (stereogram (c)); however, the enormous variation in the components removed between individual steps (stereogram (d)) makes clear identification of components difficult.

rocks of the Dokhan Volcanic Formation and Younger Granites document a prolonged history of remagnetization. Since it appears that none of the magnetic components discussed is primary, it is difficult, and perhaps somewhat premature, to relate them to specific events. Accomplishing this will require further studies, especially of the Younger Granites and their contacts. In the interpretation presented here, we record a post-granite emplacement, secondary magnetization of the Dokhan Volcanic Formation dating around 500–550 Ma, and possible late Mesozoic (AB component) and Cenozoic overprinting (C component and dike swarms) of the magnetization of the xenolithic and granitic rocks. This, in turn, suggests that even though the events which resulted in remagnetization may be general, the process at the sample level must depend closely on local conditions, for a few samples still retain a memory of earlier events (the D component).

On the basis of our data, we cannot rule out the alternative possibility that the granitic magnetization

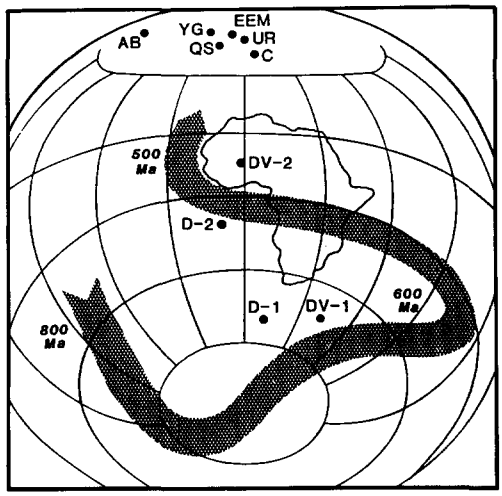



Fig. 12. The geomagnetic polar wander path for Late Precambrian–Early Paleozoic Africa from McWilliams and Kroner (1981), to which data from the Red Sea Hills has been added. QS and UR, Qena–Safaga and Um Rus dike swarms (Davies *et al.* 1980); EEM, Esh El Mellaha dikes (Abdullah *et al.* 1984); DV-1 and DV-2, Dokhan Volcanics with and without present dip correction (Perry 1982); YG, Safaga Younger Granites (this paper); AB and C, components of the xenolith and granite contact (this paper); D-1 and D-2, xenolith component with and without dip correction (this paper).

might have been generated during late phase deuteric alteration, but this would not only imply extraordinarily rapid looping movements during the Cambrian to give a pole close to the present pole, for which there is no evidence, but it would also require acceptance of the older age for the Dokhan Volcanic Formation.

*Acknowledgements*—The fieldwork for this study was carried out under the aegis of NSF Int. grant 78-01469 to Dr W. H. Kanes, University of South Carolina. The assistance in the field of Dr H. Abu Zeid and W. Smith is gratefully acknowledged. Dr R. E. Dooley is also thanked for the K/Ar determinations. Dr R. Sedivy made available his data from four sites in the granites of the Safaga region. This paper was presented in part at the IAGA meeting, Prague, 1985.

The constructive comments of the referees were invaluable in improving the manuscript and they are gratefully acknowledged.

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